

# Signal Processing: Methods and Algorithms

## Laboratory 4

### 1 Estimation of WSS random processes

This section deals with the estimation of the mean, variance, and autocorrelation function of Wide Sense Stationary (WSS) random processes. Do the following:

1. Generate  $N = 1000$  samples of a white Gaussian noise  $F[n]$  with zero mean and unit variance, then:
  - 1.1 Estimate the mean value  $\mu_F$ . **Hint:** Use the Matlab function `mean`.
  - 1.2 Estimate the variance  $\sigma_F^2$ . **Hint:** Use the Matlab function `var`.
  - 1.3 Estimate the autocorrelation function  $r_F[l]$ , both with the unbiased and biased estimators. **Hint:** Use the Matlab function `xcorr`, which implements both estimators, as described in the corresponding help.
  - 1.4 Plot the estimated unbiased and biased autocorrelation functions in the same figure. The x-axis must be the lag  $l$  ranging from  $-(N - 1)$  to  $N - 1$ .
2. Generate one realization of the Ornstein-Uhlenbeck process  $V[n]$ , by using the code written for the first laboratory. Discard the transient of the process by first simulating  $N_0 = N_{tr} + N$  samples, and then by eliminating the first  $N_{tr}$  samples, where  $N_{tr} = \frac{10}{\beta T_s}$ , and  $T_s$  is the sampling time. ( $1/\beta$  is the time constant of the Langevin equation.) With the simulated realization  $V[n]$ , do the following:
  - 2.1 Estimate the mean value  $\mu_V$ .
  - 2.2 Estimate the variance  $\sigma_V^2$ .
  - 2.3 Estimate the autocorrelation  $r_V[l]$  and compare it against the theoretical one by plotting them in the same figure. Remember that the theoretical autocorrelation function of the Ornstein-Uhlenbeck process is

$$r_V[l] = \frac{1}{2\beta} e^{-\beta |T_s| |l|} \quad (1)$$

**Hints:** Simulate a long realization of  $V[n]$  with  $N = 10^5$  samples (or, better, with  $N = 10^6$  samples, depending on the speed of your computer). Estimate the autocorrelation with a maximum lag much smaller than  $N - 1$ , to reduce the computational time. Read the Matlab help for `xcorr` to understand how to fix a maximum lag.

### 2 Estimation of nonstationary random processes

This section requires the implementation of the sliding estimators of the mean value of a nonstationary random process. Do the following:

1. Write a Matlab function that implements the sliding estimator  $\hat{\mu}_X[n]$  of the mean of a nonstationary random process  $X[n]$ . The function must have the following syntax:

$$[\text{mu\_est}] = \text{sliding\_mean}(\mathbf{x}, \text{Nw})$$

where  $\mathbf{x}$  is a vector of  $N$  elements,  $\text{Nw}$  is an even integer, smaller than  $N$ , which represents the length of the analysis window, and  $\text{mu\_est}$  is the vector of the estimates  $\hat{\mu}_X[n]$ . Remember that at the beginning and at

the end of the signal the number of available samples of the signal is smaller than  $N_w$ . **Hint:** Define the lower and upper limit of the sliding window  $N_w$  as

$$n_1 = n - N_w/2 \quad (2)$$

$$n_2 = n + N_w/2 - 1 \quad (3)$$

when  $n_1 < 0$ , that is, we are at the beginning of the signal  $X[n]$ , set  $n_1 = 0$ . When  $n_2 > N - 1$ , that is, we are at the end of the signal  $X[n]$ , set  $n_2 = N - 1$ . In the Matlab language, the windowed signal  $\mathbf{Xn}$  is given by

$$\mathbf{Xn} = \mathbf{X}(\mathbf{n1} + 1 : \mathbf{n2} + 1) \quad (4)$$

because Matlab indices go from 1 to  $N$ . Then, apply the sample mean estimator, or any other estimator, to  $\mathbf{Xn}$ .

2. Simulate one realization of a white Gaussian noise  $F[n]$  with zero mean and unit variance. Then:
  - 2.1 Obtain the estimate  $\hat{\mu}_F[n]$  of the mean value  $\mu_F[n]$  by applying the `sliding_mean` function to  $F[n]$ .
  - 2.2 Plot the estimated mean value  $\hat{\mu}_F[n]$ , the true value  $\mu_F[n]$ , and  $F[n]$  in the same figure.
3. Simulate the three nonstationary random processes with time-varying mean value and constant variance generated in Sects. 2.1-2.3 of the second laboratory. Then:
  - 3.1 Obtain the estimate  $\hat{\mu}_X[n]$  of the mean value  $\mu_X[n]$  by applying the `sliding_mean` function to  $X[n]$  for each of the three random processes.
  - 3.2 Plot the estimated mean value  $\hat{\mu}_X[n]$ , the true value  $\mu_X[n]$ , and  $X[n]$  in the same figure. Repeat the procedure for each of the three random processes.
  - 3.3 Consider the process  $X[n]$  whose mean value changes suddenly in time, and estimate  $\hat{\mu}_X[n]$  with different window lengths  $N_w$  to highlight the tradeoff between localization of events in time and variance of the estimate.